STAT201 · STATISTICS · LECTURE 1

Data, Descriptive Statistics and Visualization

Which numbers and which graphs you need to describe a data set honestly

Syllabus topic 1 · week 1 (sessions 1 and 2) · 4 academic hours

LECTURE GOALS

- determine the measurement scale of a variable and know which statistics are admissible for it;
- compute the mean, median and mode by hand and choose a suitable measure of centre;
- compute the range, variance, standard deviation and coefficient of variation and interpret them in the units of the data;
- find quartiles, the interquartile range and the five-number summary;
- build a frequency table, histogram, box plot and scatter plot and read them correctly;
- apply the $1.5 \times$ IQR rule to find outliers and report honestly what was done with them;
- tell an honest graph from a misleading one.

1 Why we need statistics and how to use these notes

Statistics is the science of drawing well-founded conclusions from data when the outcome is uncertain. A bank decides whether to grant a loan; a coffee-shop chain decides where to open a new branch; a ministry announces "the average household income". Behind each decision stand numbers, and the question is always the same: *what do these numbers really say, and what do they not say?*

The course has four parts, and this week's lecture serves as a vocabulary for the other three.

Weeks 1–2	**Describe**	What do the data contain, and how likely is a given outcome?
Weeks 3–7	**Model**	Probability distributions, expected value, the central limit theorem.
Weeks 9–11	**Infer**	Confidence intervals, hypothesis tests, p-values.
Weeks 12–15	**Predict**	Regression and the difference between correlation and causation.

How to use these notes. Read a section through, then close the text and recompute the worked example by hand without peeking. Only then move on to the problems at the end of the lecture. The glossary tables also give the Russian term, because the course also runs in Russian. The rule of the semester: **every number we compute, we must be able to explain in one sentence.**

CONVENTIONS OF THIS COURSE

Decimals are written with a point (35.75, not 35,75): this is how calculators, Excel and R print them. An answer without units is not an answer: “35.75” means nothing, “35.75 million UZS per month per branch” does.

2 Data and measurement scales

2.1 Unit of observation, variable, population and sample

BASIC CONCEPTS

Unit of observation — the object about which data are collected (a coffee shop, a customer, a region, a month). **Variable** — a characteristic of a unit that can take different values (revenue, age, region). **Population** — all the units about which we want to draw a conclusion; its size is denoted N. **Sample** — the part of the population that we actually measured; its size is denoted n.

Statistics distinguishes between population *parameters* and sample *statistics*. Parameters are written with Greek letters (μ — population mean, σ — its standard deviation), sample statistics with Latin letters ($\bar{x}$, s). The formulas are often the same and only the letter changes, but the letter tells the reader whether you measured everyone or only a part. Why this matters we shall see in weeks 7 and 9.

2.2 Four measurement scales

Not every number in a table is a number in the mathematical sense. The scale on which a variable is measured determines which calculations with it are meaningful.

Scale	What it means	Example	Admissible measure of centre
Nominal	Names only, no order	Region, industry, bank name	Mode
Ordinal	There is an order, distances between values are unequal	Credit rating, satisfaction score 1–5	Mode, median
Interval	Distances are equal, no true zero	Temperature in °C, calendar year	Mode, median, mean
Ratio	Distances are equal and there is a true zero	Revenue, price, hours worked	All three; ratios are also meaningful (“twice as much”)

COMMON MISTAKE

A number written in a table is not yet a quantitative variable. The postal code 100084 is a nominal variable: its mean makes no sense. The same holds for an employee number or a region code. A test question: “Does it make sense to add two such values?”

Likert-scale scores (“1 — strongly disagree, …, 5 — strongly agree”) are formally ordinal. In practice the mean is often computed for them, but this is an assumption and it must be stated aloud.

Quantitative variables are further divided into *discrete* (the result of counting: the number of customers per day, 0, 1, 2, …) and *continuous* (the result of measuring: revenue, weight, waiting time).

2.3 Three forms of data

- **Cross-section data**: many units at one point in time. Example: the revenue of 40 coffee shops in September 2026. We assume this form in weeks 1–11.
- **Time series**: one unit at many points in time. Example: monthly inflation in Uzbekistan over 2015–2026. The order of observations matters, so observations are not independent.
- **Panel data**: many units and many periods. Example: all 14 regions, every year for ten years. This is the main object of the econometrics course ECMT301.

3 Running example: twelve coffee shops

We illustrate all the material of the lecture with the same data. These are the monthly revenues of twelve branches of one coffee-shop chain in Tashkent for August 2026, in millions of UZS. The data are already sorted in ascending order; branch 12 is located in a shopping mall.

Number	1	2	3	4	5	6	7	8	9	10	11	12
Revenue x_i	18	22	25	27	28	30	30	33	36	40	44	**96**

Here $n = 12$, the sum is $\sum x_i = 429$, the smallest value is 18, the largest 96, and the scale is ratio. The unit of observation is a branch, the variable is monthly revenue.

4 Measures of centre

A measure of centre answers the question: “which single number best describes the typical value?” There are three such numbers, and they answer it differently.

4.1 Arithmetic mean

SAMPLE MEAN

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{1}{n}\sum_{i=1}^{n} x_i. \tag{4.1}$$

For a population the same formula gives μ, with the sum running over all N units.

EXAMPLE 1. AVERAGE BRANCH REVENUE

$\bar{x} = \dfrac{429}{12} = 35.75$ million UZS per branch per month.

The mean is the *balance point* of the data: if you picture the values as weights on a ruler, the mean is the point at which the ruler balances. This is written as an exact property:

$$\sum_{i=1}^{n}(x_i - \bar{x}) = \sum x_i - n\bar{x} = n\bar{x} - n\bar{x} = 0. \tag{4.2}$$

Deviations on one side of the mean exactly offset the deviations on the other. In our data the deviation of branch 12 is $96 - 35.75 = +60.25$, and the deviations of the other eleven branches sum to -60.25: one large value balances all the others, and that is exactly why it pulls the mean so hard.

4.2 Median

MEDIAN

The value that splits an ordered series in half: half of the observations are no larger, half are no smaller. Algorithm:

1. sort the values in ascending order;
2. find the position $(n+1)/2$;
3. if n is odd, the median is the value in that position; if n is even, the median is the average of the two values on either side of the position.

EXAMPLE 2. MEDIAN REVENUE

For $n = 12$ the position is $(12+1)/2 = 6.5$ — between the sixth and seventh values. Both equal 30, so $\text{Me} = (30+30)/2 = 30.0$ million UZS. Half of the branches earn less than 30 million, half earn more. The median ignores *how much* more, which is exactly why branch 12 does not move it.

4.3 Mode

MODE

The most frequently occurring value. For continuous data one speaks of the modal interval — the tallest bar of the histogram. There may be several modes, or none at all.

In our data the value 30 occurs twice and the rest once each, so $\text{Mo} = 30$. The mode is the only measure of centre admissible for nominal data: “the most frequent region of customers”.

4.4 Weighted mean

If the values have different importance (weight), the ordinary mean is misleading.

$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}. \tag{4.3}$$

EXAMPLE 3. AVERAGE PRICE OF A LOAF

Three shops sold 200, 500 and 300 loaves in a week at prices of 4 000, 4 500 and 5 000 UZS. The simple average of the prices is $4\,500$, but it ignores sales volumes. The weighted mean with quantities as weights:

$$\bar{x}_w = \frac{200 \cdot 4000 + 500 \cdot 4500 + 300 \cdot 5000}{200 + 500 + 300} = \frac{4\,550\,000}{1\,000} = 4\,550 \text{ UZS}.$$

This is the actual average price paid by a customer.

4.5 Outliers and the choice of a measure of centre

What happens if we exclude branch 12 from the shopping mall?

	Mean	Median	Mode
All 12 branches	35.75	30.0	30
Without branch 12 ($n = 11$)	30.27	30.0	30
Change	-5.48	**0.0**	**0**

One observation out of twelve moved the mean by 5.48 million UZS (15% of its value), while the median did not change. A statistic that barely reacts to extreme values is called **robust**. The median is robust, the mean is not. Neither is "wrong", but reporting the mean of 35.75 would describe a chain to which none of the other eleven branches belongs.

Use	When
Mean	The distribution is roughly symmetric and you need to add up or forecast totals (total revenue $= n\bar{x}$).
Median	The data are skewed or contain outliers: incomes, house prices, waiting times.
Mode	The variable is categorical or you need the most frequent option.

COMMON MISTAKE

The mean need not resemble any real observation: 35.75 is the revenue of none of the twelve branches. The "average person" often does not exist.

5 Measures of spread

One number for the centre is not enough. Consider two stalls with the same average weekly revenue of 30 million UZS:

	Weekly sales	s
Stall A	29, 30, 30, 31, 30	0.71
Stall B	5, 55, 12, 48, 30	21.78

At stall A staffing and inventory plan themselves; at stall B every week is a gamble. The mean and median are the same, the only difference is the spread.

5.1 Range

$$R = x_{\max} - x_{\min} = 96 - 18 = 78.$$

The range uses only two observations, so a single error in the data ruins it. It is worth reporting, but should not be relied on.

5.2 Variance and standard deviation

A natural idea: average the deviations from the mean. But by formula (4.2) they sum to zero. To keep the deviations from cancelling each other, they are squared.

SAMPLE VARIANCE AND STANDARD DEVIATION

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}, \qquad s = \sqrt{s^2}. \tag{5.1}$$

The variance is measured in squared units of the data ("million UZS squared") and is an intermediate step, not a result. The standard deviation s returns to the original units. It is read as follows: a typical branch lies about s away from the mean.

EXAMPLE 4. FIVE BRANCHES BY HAND

The number of customers per day at one branch: 20, 25, 30, 35, 40. The mean is $\bar{x} = 150/5 = 30$.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
20	-10	100
25	-5	25
30	0	0
35	$+5$	25
40	$+10$	100
$\sum = 150$	$\sum = 0$ (always)	$\sum = 250$

$$s^2 = \frac{250}{5-1} = 62.5, \quad s = \sqrt{62.5} = 7.91 \text{ customers.}$$

For the twelve coffee shops, computing from the definition is inconvenient, so we use the working formula, equivalent to (5.1):

$$s^2 = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}. \tag{5.2}$$

Here $\sum x_i^2 = 19\,883$ and $n\bar{x}^2 = 12 \cdot 35.75^2 = 15\,336.75$, so

$$s^2 = \frac{19\,883 - 15\,336.75}{11} = \frac{4\,546.25}{11} = 413.30, \qquad s = \sqrt{413.30} = 20.33 \text{ million UZS.}$$

Why $n-1$ in the denominator and not n

The mean $\bar{x}$ is computed from the same sample, so the data are on average closer to $\bar{x}$ than to the unknown μ, and dividing by n systematically understates the spread. In addition, by formula (4.2) the deviations from $\bar{x}$ are tied by one condition (they sum to zero), so only $n-1$ of them vary freely. The number $n-1$ is called the number of *degrees of freedom*.

EXAMPLE 5. A CHECK ON A TINY POPULATION

Let the population consist of two equally likely values, 0 and 2. Its variance is $\sigma^2 = 1$. Take samples of two values with replacement: there are four and they are equally likely.

Sample	$\bar{x}$	$\sum(x_i-\bar{x})^2$
(0, 0)	0	0
(0, 2)	1	2
(2, 0)	1	2
(2, 2)	2	0

The average of $\sum(x_i-\bar{x})^2$ over the four samples is $(0+2+2+0)/4 = 1$. Dividing by $n-1=1$ gives 1 on average — correct, equal to σ^2. Dividing by $n=2$ gives 0.5 on average — half of the truth. Conclusion: with $n-1$ the sample variance is an *unbiased* estimator. More on this in week 9.

5.3 Coefficient of variation

Standard deviations cannot be compared if the data are measured in different units or on different scales. A relative measure serves this purpose:

$$CV = \frac{s}{\bar{x}} \times 100\%. \tag{5.3}$$

EXAMPLE 6. COFFEE SHOPS AND A BANK

Coffee-shop chain: $\bar{x} = 35.75$, $s = 20.33$, so $CV = 56.9\%$. Bank branches: $\bar{x} = 1\,240$ million, $s = 186$ million, so $CV = 15.0\%$. The bank's standard deviation in absolute money is nine times larger, but relative to their own scale its branches are far more homogeneous.

COMMON MISTAKE

Never compare two standard deviations measured in different units: compare coefficients of variation. But CV makes sense only for ratio-scale variables with a positive mean (for temperature in °C, for example, it is meaningless: the zero is arbitrary).

5.4 What happens to the measures under a data transformation

Let each value be recomputed by the rule $y_i = a + b\,x_i$ (change of units, a bonus, indexation).

LINEAR TRANSFORMATION

$$\bar{y} = a + b\bar{x}, \qquad \mathrm{Me}_y = a + b\,\mathrm{Me}_x, \qquad s_y = |b|\,s_x.$$

Adding a constant a shifts the centre but **does not change the spread**. Multiplying by b stretches both the centre and the spread. The coefficient of variation does not change under multiplication ($a = 0$) but does change under a shift.

6 Measures of position: quartiles and the five-number summary

Quartiles split ordered data into four parts with equal numbers of observations. Q_2 is the median; Q_1 is the median of the lower half; Q_3 is the median of the upper half. If n is odd, the median itself belongs to neither half.

EXAMPLE 7. QUARTILES OF THE COFFEE SHOPS

Lower half: 18, 22, 25, 27, 28, 30. Its median is $Q_1 = (25 + 27)/2 = 26$.
Upper half: 30, 33, 36, 40, 44, 96. Its median is $Q_3 = (36 + 40)/2 = 38$.
Interquartile range $\mathrm{IQR} = Q_3 - Q_1 = 38 - 26 = 12$.

FIVE-NUMBER SUMMARY

$$x_{\min}, \quad Q_1, \quad \mathrm{Me}, \quad Q_3, \quad x_{\max} \qquad \Rightarrow \qquad 18,\ 26,\ 30,\ 38,\ 96.$$

The middle 50% of branches earn between 26 and 38 million UZS, i.e. lie in a band 12 million wide; compare this with the range of 78. These numbers alone show that one observation behaves unusually.

QUARTILES IN SOFTWARE

There are several ways to compute quartiles and they give slightly different numbers. For our data: the halves method (used in this course) gives $Q_1 = 26$ and $Q_3 = 38$; the Excel function `QUARTILE.INC` and R by default give 26.5 and 37; the function `QUARTILE.EXC` gives 25.5 and 39. For large n the difference is small. On paper in this course we use the halves method, and when working with software we always state which method was used.

A *percentile* generalizes the quartile: the p-th percentile is the value below which $p\%$ of the observations lie. Quartiles are the 25th, 50th and 75th percentiles.

7 Frequency distributions and histograms

When there are many observations, a list of values is unreadable and the data are grouped into intervals.

Construction algorithm. (1) Choose the interval width. (2) Define intervals closed on the left: $[15, 25)$ includes 15 but not 25. (3) Count the observations in each interval — the *frequency* f. (4) Find the *relative frequency* f/n and the *cumulative frequency*.

Interval, million UZS	Frequency f	Rel. frequency	Cumulative
[15, 25)	2	16.7%	2
[25, 35)	6	50.0%	8
[35, 45)	3	25.0%	11
[45, 55)	0	0.0%	11
[55, 65)	0	0.0%	11
[65, 75)	0	0.0%	11
[75, 85)	0	0.0%	11
[85, 95)	0	0.0%	11
[95, 105)	1	8.3%	12
Total	**12**	**100%**	

Check: the frequencies sum to $n = 12$, the relative frequencies to 100%.

A **histogram** is the graph of this table: intervals along the horizontal axis, frequencies along the vertical. The bars touch because revenue is continuous. The main decision in constructing it is the interval width.

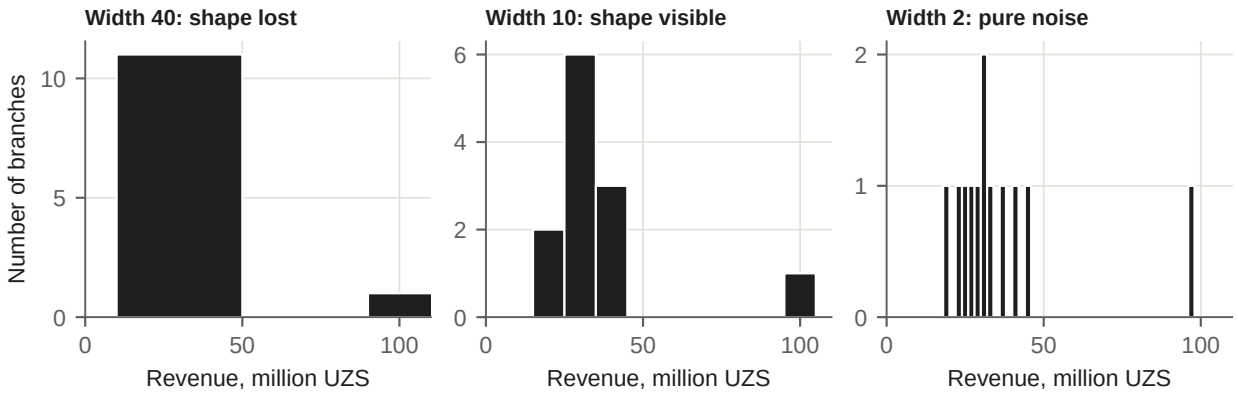


Fig. 1. The same 12 values with different interval widths. Intervals that are too wide erase the shape, intervals that are too narrow leave noise.

Rule of thumb: start with a number of intervals close to $\sqrt{n}$ or to $1 + \log_2 n$ (Sturges' rule) and adjust until the shape stops changing; always state the interval width in the caption. For $n = 12$ any histogram is crude; the rule works well from about $n \approx 50$. In the figure, with width 10 you can clearly see the core (25–45 million), the long right tail and the empty gap before the lone bar at 96.

8 Shape of a distribution

Shape describes symmetry and tails. Three main situations:

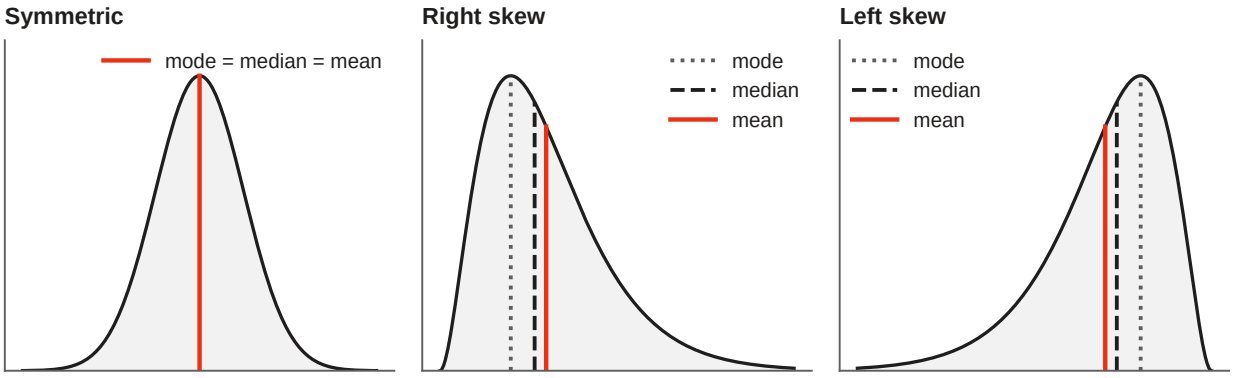


Fig. 2. Positions of the mode, median and mean for three distribution shapes.

- **Symmetric**: mean = median = mode. Human height, measurement errors, a well-designed exam.
- **Right-skewed** (long tail to the right): mode < median < mean. Incomes, house prices, waiting times, our coffee-shop chain.
- **Left-skewed** (long tail to the left): mean < median < mode. Age at death, scores on an easy test.

The mean is "dragged" along with the tail while the median hardly moves — so the order of mode, median and mean hints at the shape.

Skewness can be measured by a number. The simplest coefficient (Pearson's second coefficient):

$$Sk = \frac{3(\bar{x} - \mathrm{Me})}{s}. \tag{8.1}$$

For the coffee shops: $Sk = \dfrac{3(35.75 - 30)}{20.33} = \dfrac{17.25}{20.33} = +0.85$. Guidelines: $Sk \approx 0$ — symmetry; from 0.5 to 1 — moderate right skew; above 1 — strong; a negative value — tail to the left. Software usually reports a different coefficient (based on the third moment); its sign is the same.

9 The box plot and outliers

A **box plot** draws the five-number summary: a box from Q_1 to Q_3, a line inside the box at the median (not the mean), "whiskers" extending to the most distant observations that do not go beyond the fences, and observations beyond the fences marked as separate points.

THE $1.5 \times$ IQR RULE

$$\text{lower fence} = Q_1 - 1.5 \cdot \mathrm{IQR}, \tag{9.1}$$

$$\text{upper fence} = Q_3 + 1.5 \cdot \mathrm{IQR}. \tag{9.2}$$

An observation beyond the fences is called an **outlier**.

EXAMPLE 8. IS BRANCH 12 AN OUTLIER?

$\mathrm{IQR} = 12$, so $1.5 \cdot 12 = 18$. The fences are $26 - 18 = \mathbf{8}$ and $38 + 18 = \mathbf{56}$. Since $96 > 56$, branch 12 is an outlier. The whiskers go to 18 and 44 — the most extreme values inside the fences.

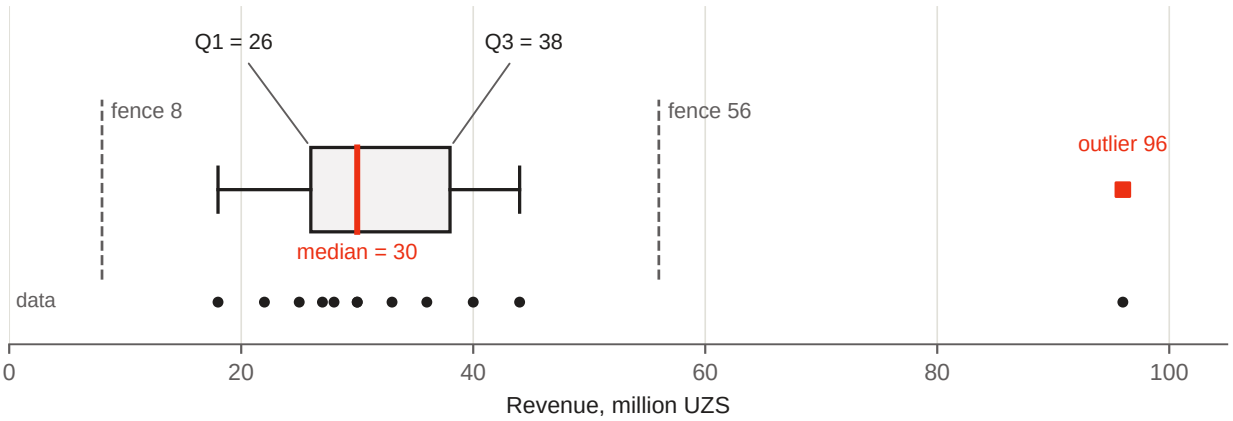


Fig. 3. Box plot of the revenue of the twelve branches. The data themselves are shown below the box.

An outlier is a definition, not an opinion. The rule gives every analyst the same answer.

9.1 What to do with an outlier

The rule only *flags* an observation for inspection; it does not say the observation is wrong. There are four admissible actions:

1. **Check**: is it a typo (96 instead of 9.6)? Is it a different unit of measurement?
2. **Explain**: a branch in a shopping mall has a different business model.
3. **Report both results**: "mean 35.75 million for all branches; 30.27 million without the mall branch".
4. **Change the statistic**: use the median and IQR.

> **AN INADMISSIBLE ACTION**
>
> Quietly deleting an outlier so the data "look better". Any exclusion of observations must be recorded in the text of the report.

9.2 Missing values

An outlier is not the only data problem. If revenue is not recorded for 3 of 40 coffee shops, you must report how many observations were actually used and think about *why* the values are missing. If the gaps are random, the loss is small. But if mostly loss-making branches are missing because they did not file their reports, the mean of the remaining ones is systematically too high. The rule is the same as for outliers: hide nothing and describe how missing values were handled.

10 The scatter plot

When each unit has two numerical variables, their relationship is shown by a **scatter plot**: one point per unit. The explanatory variable is plotted horizontally, the response vertically. Let us add to our data the branches' advertising spend x (million UZS) and revenue y (million UZS):

Branch	1	2	3	4	5	6	7	8	9	10	11	12
Advertising x	1.0	1.4	1.6	1.9	2.0	2.3	2.4	2.8	3.1	3.5	3.9	5.6
Revenue y	18	22	25	27	28	30	30	33	36	40	44	96

(Illustrative data.)

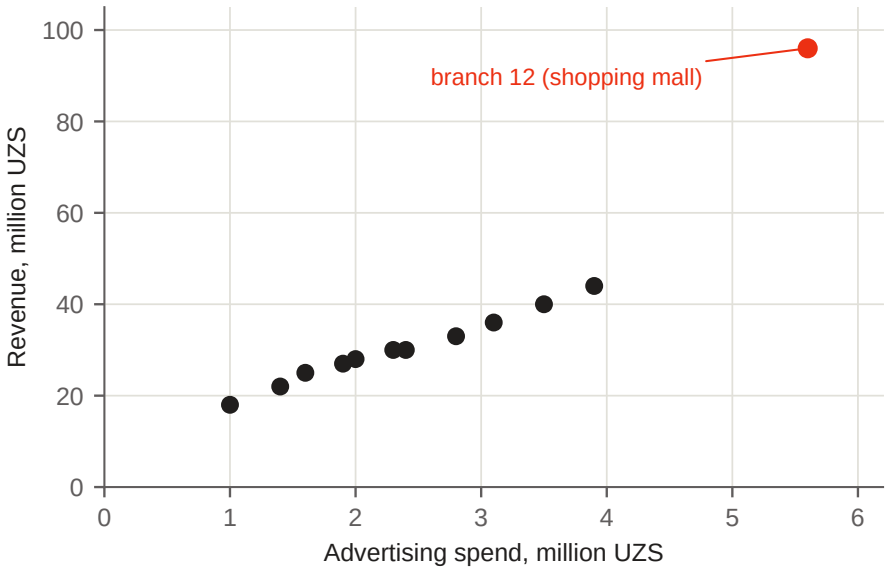


Fig. 4. Advertising spend and revenue of the twelve branches.

Describing a scatter plot, one speaks of four things:

- **Direction**: positive (as advertising grows, revenue grows).
- **Form**: roughly linear. A linear model is justified only for a straight band of points.
- **Strength**: strong. In week 6 we shall learn to measure it by a number — the correlation coefficient r (for these data $r = 0.93$, and without branch 12 it is $r = 0.996$).
- **Unusual points**: branch 12.

> **COMMON MISTAKE**
>
> That advertising and revenue grow together does not prove that advertising *causes* revenue to grow: perhaps large branches simply spend more on advertising. We shall examine the difference between correlation and causation in week 13.

11 How to choose a graph and not deceive the reader

What you have	Which graph	Why
One categorical variable	Bar chart	Length is the easiest thing to compare.
One numerical variable	Histogram or box plot	Shape and outliers are visible at once.
One numerical variable, several groups	Side-by-side box plots	Centre, spread and shape of the groups in one frame.
Two numerical variables	Scatter plot	You see the joint picture, not two separate summaries.
A numerical variable over time	Line chart	Order matters and the line conveys it.
Parts of a whole	Still a bar chart	Angles are hard to read; a pie chart is unsuitable with more than three shares.

11.1 Honest numbers, dishonest picture

- **A truncated axis.** If the y-axis does not start at zero, a small increase looks like a cliff. Bar charts must start at zero.
- **Area instead of length.** Doubling both height and width gives four times the "ink" for a doubled value.
- **A pie chart with seven shares.** Nobody can rank the shares by size; a sorted bar chart answers that question at once.
- **Two axes on one graph.** By choosing the scales one can make lines cross and create a relationship that does not exist.

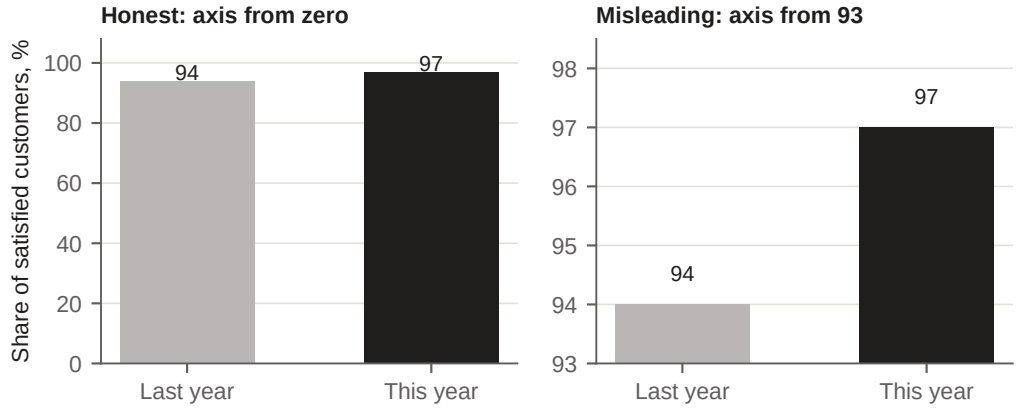


Fig. 5. The same data: the share of satisfied customers rose from 94% to 97%, i.e. by 3.2%. In the right-hand graph this year's bar looks four times taller than last year's.

A **graph caption** states the conclusion, not the type of graph. Bad: "Box plot of revenue". Good: "One mall branch earns three times the median".

12 Key terms

Term	*In Russian*	Meaning
Population	*Генеральная совокупность*	all units about which the conclusion is drawn (N, μ, σ)
Sample	*Выборка*	the measured part of the population (n, $\bar{x}$, s)
Scale of measurement	*Шкала измерения*	nominal, ordinal, interval, ratio
Mean	*Среднее*	the sum divided by the number of values; the balance point
Median	*Медиана*	the middle of the ordered series
Mode	*Мода*	the most frequent value
Robust	*Устойчивый*	hardly dependent on extreme values
Range	*Размах*	$x_{\max} - x_{\min}$
Variance	*Дисперсия*	the sum of squared deviations divided by $n-1$
Standard deviation	*Стандартное отклонение, СКО*	the square root of the variance, in the units of the data
Coefficient of variation	*Коэффициент вариации*	$s/\bar{x}$, a dimensionless relative measure of spread
Quartile	*Квартиль*	points dividing the data into quarters
IQR (interquartile range)	*Межквартильный размах*	$Q_3 - Q_1$
Skewness	*Асимметрия*	lack of symmetry of a distribution
Box plot	*Ящик с усами*	graph of the five-number summary
Outlier, fence	*Выброс, граница выброса*	a value beyond $Q_1 - 1.5\text{IQR}$, $Q_3 + 1.5\text{IQR}$
Scatter plot	*Диаграмма рассеяния*	points (x_i, y_i) for two variables
Histogram	*Гистограмма*	graph of frequencies by interval

Formulas of Lecture 1

$$\bar{x} = \frac{1}{n}\sum x_i, \qquad \bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}, \qquad \sum (x_i - \bar{x}) = 0,$$

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}, \qquad s = \sqrt{s^2}, \qquad CV = \frac{s}{\bar{x}} \cdot 100\%,$$

$$R = x_{\max} - x_{\min}, \qquad \text{IQR} = Q_3 - Q_1, \qquad Sk = \frac{3(\bar{x} - \text{Me})}{s},$$

outlier fences: $Q_1 - 1.5\,\text{IQR}$, $Q_3 + 1.5\,\text{IQR}$,

$y = a + bx \;\Rightarrow\; \bar{y} = a + b\bar{x}, \;\; s_y = |b|\, s_x.$

13 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix "Statistical Tables".

Problem 1 [A] For each variable state the measurement scale and which measures of centre (mode, median, mean) are admissible:

(a) employee number;
(b) years of education;
(c) satisfaction with a service on a scale of 1–5;
(d) monthly salary in UZS;
(e) region of residence;
(f) year of birth;
(g) air temperature in °C;
(h) place achieved in a competition (1st, 2nd, …).

Problem 2 [A] True or false? Justify briefly.

(a) The median does not depend on the size of the extreme values.
(b) If the mean is greater than the median, the distribution is, as a rule, right-skewed.
(c) The sum of deviations from the median is always zero.
(d) The variance can be negative.
(e) If all values are multiplied by 2, the standard deviation doubles and the coefficient of variation does not change.
(f) The line inside the box of a box plot shows the mean.

Problem 3 [B] A courier's number of deliveries over seven days: 12, 15, 15, 18, 22, 25, 61. Find the mean, median, mode, range, variance, standard deviation and coefficient of variation. Use the $1.5 \times$ IQR rule to check for outliers. Which measure of centre would you publish and why? Answer in one sentence.

Problem 4 [B]

(a) A bank issued three loans: 50 million UZS at 18%, 30 million at 22% and 20 million at 24%. Find the average interest rate of the portfolio (simple and weighted) and explain the difference.
(b) A student got 85 points in a 5-credit course, 70 points in a 4-credit course and 90 points in a 3-credit course. Find the weighted average score.

Problem 5 [B] Monthly rent of 15 apartments (million UZS): 2.4, 2.6, 2.8, 3.0, 3.1, 3.2, 3.5, 3.6, 3.8, 4.0, 4.2, 4.5, 4.8, 5.0, 9.5.

(a) Find the five-number summary and the IQR.
(b) Find the outlier fences and identify the outliers.
(c) Describe the box plot: where do the whiskers end?

(d) Compare the mean and the median. What do they say about the shape of the distribution? Which measure would you report?

Problem 6 [B] Number of orders at a coffee shop over 30 days: 14, 18, 19, 21, 22, 23, 24, 24, 25, 26, 27, 27, 28, 29, 29, 30, 31, 31, 32, 33, 34, 35, 36, 37, 38, 40, 42, 45, 49, 58.

(a) Build a frequency table with intervals $[10, 20), [20, 30), \dots, [50, 60)$: frequency, relative frequency, cumulative frequency.
(b) On what share of days were there fewer than 30 orders?
(c) State the modal interval.
(d) Find the mean and median from the raw data and compare them. What does this say about the shape?

Problem 7 [B]

(a) The average salary in firm A is 2.4 million UZS with a standard deviation of 0.6 million; in firm B it is 5.0 million with a standard deviation of 1.0 million. Where are salaries more homogeneous?
(b) The return on stock X: mean 8%, standard deviation 12%; on stock Y: mean 5%, standard deviation 6%. Which stock has less risk per unit of return?

Problem 8 [C]

(a) Prove that $\sum(x_i - \bar{x}) = 0$.
(b) The salaries of five employees (million UZS): 2.0, 2.5, 3.0, 3.5, 4.0. Find $\bar{x}$, s and CV. Then each salary is raised by 10% and a bonus of 0.2 million is added: $y = 1.1x + 0.2$. Find $\bar{y}$, s_y, CV_y and explain why CV changed.

Problem 9 [C] Monthly incomes of 10 households (million UZS): 2.1, 2.4, 2.6, 2.8, 3.0, 3.1, 3.4, 3.9, 4.5, 18.2. The ministry reports “average income 4.6 million”, the trade union “typical income 3.05 million”. Both use the same data honestly.

(a) Verify both numbers.
(b) How many of the ten households have an income below the “average”?
(c) Check for outliers.
(d) Which number should a family planning its budget use, and why?

Problem 10 [C] A company reports that the share of satisfied customers rose from 94% to 97%. In the bar chart the y-axis starts at 93%.

(a) By how many percentage points and by what relative percentage did the share grow?
(b) How many times taller does this year’s bar look than last year’s?
(c) How would you fix the graph? Write a caption for the corrected graph.

Problem 11 [C] Hours of preparation x and score y for eight students:
$(1, 45), (2, 52), (3, 58), (4, 61), (5, 70), (6, 74), (7, 79), (8, 60)$.

(a) Draw the scatter plot.
(b) Describe it in four words: direction, form, strength, unusual points.
(c) What will you do with the point $(8, 60)$ before drawing conclusions?
(d) Can we claim that more hours of preparation *cause* a higher score?